

Machine 5: Decimal 64-bit Floating-Point Machine

Process: IEEE 754 decimal double-precision operations

S02 Group 5:

Doctora, Justin S

Iringan, Jamie Reign A.

Ramos, Margaret Patrice M.

Tiongco, Kyan Thomas T.

Test Cases Documentation**1. Convert decimal to decimal-based double precision***Input : (FINITE)*

9.999999999999999e384

Output :

0 11101 11111111 0011111111 0011111111 0011111111 0011111111 0011111111

77FCFF3FCFF3FCFF

Decimal-based Double Precision

9.999999999999999e384

ENCODE / DECODE**ENCODED RESULT**Type: **Finite**Hex: **0x77FCFF3FCFF3FCFF**

Binary:

0 11101 11111111 0011111111 0011111111 0011111111 0011111111 0011111111 0011111111

Input : (FINITE)

1e-383

Output :

0 00001 00000000 0000000000 0000000000 0000000000 0000000000 0000000000
040000000000000000

Decimal-based Double Precision

1e-383

ENCODE / DECODE

ENCODED RESULT

Type: **Finite**

Hex: **0x0400000000000000**

Binary:

0 00001 00000000 0000000000 0000000000 0000000000 0000
000000 0000000000

Input : (+INFINITY)

+INF

Output :

0 11110 00000000 0000000000 0000000000 0000000000 0000000000
780000000000000000

Decimal-based Double Precision

+INF

ENCODE / DECODE

ENCODED RESULT

Type: **Infinity**

Hex: **0x7800000000000000**

Binary:

0 11110 00000000 0000000000 0000000000 0000000000 0000
000000 0000000000

Input : (-INFINITY)

-INF

Output :

1 11110 00000000 0000000000 0000000000 0000000000 0000000000

F800000000000000

Decimal-based Double Precision

-INF

ENCODE / DECODE

ENCODED RESULT

Type: **Infinity**

Hex: **0xF800000000000000**

Binary:

1 11110 00000000 0000000000 0000000000 0000000000 0000
000000 0000000000

Input : Nan

Output :

0 11111 000000000 0000000000 0000000000 0000000000 0000000000
7C00000000000000

Decimal-based Double Precision

NaN

ENCODE / DECODE

ENCODED RESULT

Type: NaN

Hex: 0x7C00000000000000

Binary:

0 11111 00000000 0000000000 0000000000 0000000000 0000
000000 0000000000

Input :

-0

Output :

1 00000 00000000 0000000000 0000000000 0000000000 0000000000 0000000000
8000000000000000

Decimal-based Double Precision

-0

ENCODE / DECODE

ENCODED RESULT

Type: -0

Hex: 0x8000000000000000

Binary:

1 00000 00000000 0000000000 0000000000 0000000000 0000
000000 0000000000

Input :

+0

Output :

0 00000 00000000 0000000000 0000000000 0000000000 0000000000 0000000000
0000000000000000

Decimal-based Double Precision

+0

ENCODE / DECODE

ENCODED RESULT

Type: +0

Hex: 0x0000000000000000

Binary:

0 00000 00000000 0000000000 0000000000 0000000000 0000
000000 0000000000

2. Demonstrate rounding methods

Call:

decimalRounding("123.456789", 5)

Expected GRS:

G = 6, R = 7, S = 1

Expected results:

Truncation: 123.45 (roundedDigits: 12345, exponent: -2)

Round-Up: 123.46 (roundedDigits: 12346, exponent: -2)

Round-Down: 123.45 (roundedDigits: 12345, exponent: -2)

Nearest-Even: 123.46 (roundedDigits: 12346, exponent: -2)

ROUNDING (BASE 10)

Parsed Original: 123.456789

Truncate:

12345 (exp -2)

Round Up:

12346 (exp -2)

Round Down:

12345 (exp -2)

Round Nearest:

12346 (exp -2)

[GRS Bits: 6 7 1]

Call:

decimalRounding("-123.456789", 5)

Expected GRS:

G = 6, R = 7, S = 1

Expected results:

Truncation: -123.45

Round-Up: -123.45

Round-Down: -123.46

Nearest-Even: -123.46

ROUNDING (BASE 10)

Parsed Original: -123.456789

Truncate:

12345 (exp -2)

Round Up:

12345 (exp -2)

Round Down:

12346 (exp -2)

Round Nearest:

12346 (exp -2)

[GRS Bits: 6 7 1]

Call:

decimalRounding("12.345", 4)

Expected GRS:

G = 5, R = 0, S = 0

Expected results:

Truncation: 12.34

Round-Up: 12.35

Round-Down: 12.34

Nearest-Even: 12.34

ROUNDING (BASE 10)

Parsed Original: 12.345

Truncate:

1234 (exp -2)

Round Up:

1235 (exp -2)

Round Down:

1234 (exp -2)

Round Nearest:

1234 (exp -2)

[GRS Bits: 5 0 0]

Call:

`decimalRounding("12.355", 4)`

Expected GRS:

G = 5, R = 0, S = 0

Expected results:

Truncation: 12.35

Round-Up: 12.36

Round-Down: 12.35

Nearest-Even: 12.36

ROUNDING (BASE 10)

Parsed Original: 12.355

Truncate:

1235 (exp -2)

Round Up:

1236 (exp -2)

Round Down:

1235 (exp -2)

Round Nearest:

1236 (exp -2)

[GRS Bits: 5 0 0]

Call:

binaryRounding("101.1011", 4)

Expected GRS:

G = 0, R = 1, S = 1

Expected results:

Truncation: 101.1

Round-Up: 110.0

Round-Down: 101.1

Nearest-Even: 101.1

ROUNDING (BASE 2)

Parsed Original: 101.1011

Truncate:

1011 (exp -1)

Round Up:

1100 (exp -1)

Round Down:

1011 (exp -1)

Round Nearest:

1011 (exp -1)

[GRS Bits: 0 1 1]

Call:

binaryRounding("1111.1", 4)

Expected GRS:

G = 1, R = 0, S = 0

Expected results:

Truncation: 1111

Round-Up: 10000

Round-Down: 1111

Nearest-Even: 10000

Expected nearest-even internal result:

roundedDigits: 1000

finalExponent: 1

base: 2

ROUNDING (BASE 2)

Parsed Original: 1111.1

Truncate:

1111 (exp 0)

Round Up:

1000 (exp 1)

Round Down:

1111 (exp 0)

Round Nearest:

1000 (exp 1)

[GRS Bits: 1 0 0]

Call:

`binaryRounding("102.1", 4)`

Expected:

Error: Invalid binary input. Must be a valid binary number.

Rounding Demonstrator

102.1

4

Base 2 (Binary) ▼

APPLY ROUNDING RULES

Error: Invalid binary input. Must be a valid binary number.

Call:

```
decimalRounding("123.45", 0)
```

Expected:

Error: Invalid precision input. Must be a positive integer.

Rounding Demonstrator

123.45

0

Base 10 (Decimal) ▼

APPLY ROUNDING RULES

Error: Precision must be a positive integer.

3. Perform arithmetic operations (subtraction and division) using GRS method

Arithmetic Logic Unit

15.5

2.25

Subtraction

CALCULATE

TRACE: SUBTRACTION

A (exp -14): 1550000000000000
B (exp -15): 2250000000000000 [GRS: 000]

Result (exp -14): 1325000000000000

STEP-BY-STEP SOLUTION

i) Decimal Breakdown:

1325000000000000 $\times 10^{-14}$
Standard Value: 13.25
Significand in decimal? yes
Base-10? yes
Normalized? Yes, 16 whole digits
MSD = 1 (0001)
Sign bit = 0 (+)
 $e' = e + 398 \rightarrow -14 + 398 = 384$

ii) Binary with proper spacing:

SIGN	COMBINATION	EXP CONT.	COEFFICIENT CONT.
0	01001	10000000	0110100101 0000000000 0000000000 00000000 00000000

iii)
Hexadecimal: 0x2601A50000000000

Arithmetic Logic Unit

2.25

15.5

Subtraction

CALCULATE

TRACE: SUBTRACTION

A (exp -15): 2250000000000000
B (exp -14): 1550000000000000 [GRS: 000]

Result (exp -14): -1325000000000000

STEP-BY-STEP SOLUTION

i) Decimal Breakdown:

-1325000000000000 $\times 10^{-14}$
Standard Value: -13.25
Significand in decimal? yes
Base-10? yes
Normalized? Yes, 16 whole digits
MSD = 1 (0001)
Sign bit = 1 (-)
 $e' = e + 398 \rightarrow -14 + 398 = 384$

ii) Binary with proper spacing:

SIGN	COMBINATION	EXP CONT.	COEFFICIENT CONT.
1	01001	10000000	0110100101 0000000000 0000000000 00000000 00000000

iii)
Hexadecimal: 0xA601A50000000000

Arithmetic Logic Unit

15.5

2

Division

CALCULATE

TRACE: DIVISION

Op A: 15500000000000000 x 10⁻¹⁴

Op B: 20000000000000000 x 10⁻¹⁵

Raw Coeff: 77500000000000000

Calculated Exp: -17

STEP-BY-STEP SOLUTION

i) Decimal Breakdown:
77500000000000000 x 10⁻¹⁷
Standard Value: 7.7500000000000001
Significand in decimal? yes
Base-10? yes
Normalized? Yes, 16 whole digits
MSD = 7 (0111)
Sign bit = 0 (+)
e' = e + 398 → -17 + 398 = 381

ii) Binary with proper spacing:

SIGN	COMBINATION	EXP CONT.	COEFFICIENT CONT.
0	01111	01111101	1111010000 0000000000 0000000000 000 00000000 0000000000

iii)
Hexadecimal: 0x3DF7D00000000000

Arithmetic Logic Unit

-100.0

4.0

Division

CALCULATE

TRACE: DIVISION

Op A: 10000000000000000 x 10⁻¹³

Op B: 40000000000000000 x 10⁻¹⁵

Raw Coeff: -25000000000000000

Calculated Exp: -16

STEP-BY-STEP SOLUTION

i) Decimal Breakdown:
-25000000000000000 x 10⁻¹⁶
Standard Value: -25
Significand in decimal? yes
Base-10? yes
Normalized? Yes, 16 whole digits
MSD = 2 (0010)
Sign bit = 1 (-)
e' = e + 398 → -16 + 398 = 382

ii) Binary with proper spacing:

SIGN	COMBINATION	EXP CONT.	COEFFICIENT CONT.
1	01010	01111110	1010000000 0000000000 0000000000 000 00000000 0000000000

iii)
Hexadecimal: 0xA9FA800000000000

Arithmetic Logic Unit

1.0

3.0

Division

CALCULATE

TRACE: DIVISION

Op A:1000000000000000 x 10^-15

Op B:3000000000000000 x 10^-15

Raw Coeff:3333333333333333

Calculated Exp:-18

STEP-BY-STEP SOLUTION

i) Decimal Breakdown:

3333333333333333 x 10^-18

Standard Value: 0.3333333333333333

Significand in decimal? yes

Base-10? yes

Normalized? Yes, 16 whole digits

MSD = 3 (0011)

Sign bit = 0 (+)

e' = e + 398 → -18 + 398 = 380

ii) Binary with proper spacing:

SIGN	COMBINATION	EXP CONT.	COEFFICIENT CONT.
0	01011	01111100	0110110011 0110110011 0110110011 0110110011

iii)

Hexadecimal: 0x2DF1B36CDB36CDB3

Arithmetic Logic Unit

15.5

0.0

Division

CALCULATE

TRACE: SPECIAL CASE

Arithmetic resulted in a special state: Infinity

STEP-BY-STEP SOLUTION

i) Decimal Breakdown:

Value: Infinity

Normalized? No (Infinity)

Combination field: 11110 (Infinity)

ii) Binary with proper spacing:

SIGN	COMBINATION	EXP CONT.	COEFFICIENT CONT.
0	11110	00000000	0000000000 0000000000 0000000000 0000000000

iii)

Hexadecimal: 0x7800000000000000